Book scope: The Laws of Stacking

Halves, swaps and hexagonality, from geometry to real materials (placeholder title)



DICTO · scope for decision · 28 October 2026

> **Decision proposed.** One book from the Rule of Halves (#57) to the magnets (#74): 14 papers in a prologue, two parts and an epilogue, about 80 pages, one notation, one bibliography. Not new research: the papers tightened into chapters. Four papers from the same weeks belong to other lines and stay out.

In the book (14 papers)

	chapter	from
Prologue		
0	From the octet to every stacking: how the rule was found (short, for the general reader)	#58
Part I. The laws		
1	The Rule of Halves: the holes of every close-packed stacking	#57
2	DICTO Janus c/h Pyrochlore: the halves on the pyrochlore network	#60
3	The Swap Rule: Ganesha, the Janus tetrahedra, the border at close packing	#61
4	The Swap Rule in crystals: rock salt, cadmium iodide, zinc blende, Laves	#64
5	The Focus: the Hexagonality Law	#66
6	Random stackings and stacking faults	#67
7	The Fibonacci stacking: the golden ratio in the volumes	#69
Part II. Real materials		
8	Real polytypes: silicon carbide, silicon, germanium, cerium	#68
9	Entropy and pressure	#70
10	Alloys and hydrogen	#71
11	Oxides: perovskites and battery cathodes	#72
12	Ice, diamond and nanowires	#73
13	Spin ice and kagome magnets	#74
Epilogue		
14	One table of every material by its distance from the law; the open problems (the Loop Equality, the ice prediction, the 6H cell of $BaRuO_3$ ...)	new, short

Out of the book (other lines, same weeks)

#59 The Icosaxes Hexa · #62 Shadows of the Fourth Dimension · #63 The White Queen 600-cell · #65 The Red Queen 120-cell. They belong to the icosahedral and four-dimensional work, a book of their own later if wanted.

Editorial rules

(1) One notation throughout: c and h layers, hexagonality h, cubicity $1 - h$, the layer couplings J_1, J_2, J_3 as in #68. (2) Each chapter is its paper, repeated introductions cut,

theorems and proofs kept word for word, figures kept. (3) Later corrections carried in (for example, #73 says it completes the materials series; #74 followed). (4) One bibliography, each source marked read or cited as given; an index of materials and names. (5) A short opening for the general reader, as in Quanta of Space. (6) Published as its own Zenodo record (book) with a DICTO DOI, listed in Collected Works; licence CC BY-SA 4.0.

Steps

1. Assemble the chapters from the paper sources. 2. Notation and correction pass. 3. Merge the bibliography and build the index. 4. Typeset; DICTO reads. 5. Publish.